TOR AMPLITUDE OVER THE BASE IS DETECTED ON THE LOCALISATIONS OF THE COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. The author closes the proof with the sentence that the proofs of the two remaining implications are omitted; that sentence is reproduced verbatim inside the counted proof span and the omitted implications were not reconstructed here.

Allowed standard prerequisites (editorial).

- Tor amplitude of a complex over a ring, defined by the vanishing of the derived tensor product against any module outside the interval. defined earlier in the same section, source lines 17040-17098. Supplies the three conditions whose equivalence is asserted.
- Localisation of the cohomology of a derived tensor product computes the cohomology of the localised complex against the localised module. used at proof lines 17195-17199. Identifies the localised cohomology modules appearing in the argument.
- A module is zero if and only if all its localisations at maximal ideals are zero. the author cites Algebra, Lemma algebra-lemma-characterize-zero-local at proof lines 17201-17203. Concludes the vanishing that gives the first condition.

Source reference. Original lemma 0B67: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14060–14060. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00HN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 649–649. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B$ be a ring map. Let $K^\bullet$ be a complex of B-modules. Let $a, b \in \mathbf{Z}$. The following are equivalent*

(1) *$K^\bullet$ has tor amplitude in $[a, b]$ as a complex of A-modules,*
(2) *$K^\bullet_\mathfrak{q}$ has tor amplitude in $[a, b]$ as a complex of $A_\mathfrak{p}$-modules for every prime $\mathfrak{q} \subset B$ with $\mathfrak{p} = A \cap \mathfrak{q}$,*
of $A_\mathfrak{p}$-modules for every maximal ideal $\mathfrak{m} \subset B$ with $\mathfrak{p} = A \cap \mathfrak{m}$.

Proof. Assume (3) and let M be an A-module. Then $H^i = H^i(K^\bullet \otimes^{\mathbf{L}}_A M)$ is a B-module and $(H^i)_\mathfrak{m} = H^i(K^\bullet_\mathfrak{m} \otimes^{\mathbf{L}}_{A_\mathfrak{p}} M_\mathfrak{p})$. Hence $H^i = 0$ for $i \notin [a, b]$ by Algebra, Lemma 00HN. Thus (3) $\Rightarrow$ (1). We omit the proofs of (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (3). $\square$